White Accumulation: a claim-by-claim novelty audit

The White Accumulation framework advances eight interlocking claims about consciousness, time, and physics. **None of the eight claims has a direct published precedent that captures all of its elements simultaneously**, but the degree of prior art varies enormously — from Claim 7 (anesthesia), where the empirical content overlaps ~80% with existing work and the novelty is primarily terminological, to Claims 2 and 6, where no close match exists in the published literature. Below, each claim is examined against the closest existing work, with precise citations and an honest assessment of what is borrowed, what is recombined, and what is new.

Claim 1: Consciousness lives in trajectories, not states

The claim. Consciousness is not a property of instantaneous states but of temporally extended trajectories satisfying a global self-consistency constraint across time.

(a) Closest existing work. The strongest precedent is **Fekete and Edelman's Dynamical Emergence Theory (DET)**: Fekete, T. & Edelman, S. (2011), "Towards a Computational Theory of Experience," *Consciousness and Cognition* 20(3), 807–827; and Moyal, R., Fekete, T. & Edelman, S. (2020), "Dynamical Emergence Theory (DET): A Computational Account of Phenomenal Consciousness," *Minds and Machines* 30(1), 1–21. DET explicitly argues that "experience is realized by dynamical activity-space trajectories (rather than points)" PhilPapers and that "a time-frozen snapshot of a system's instantaneous state needs an external interpreter to qualify as a quale." Shimon-edelman An experience corresponds to "a temporally extended dynamical transient" spanning hundreds of milliseconds to seconds. Consciousnessrealist

Further precedents include **Francisco Varela's** neurophenomenology — Varela, F. (1999), "The Specious Present: A Neurophenomenology of Time Consciousness," in Petitot et al. (eds.), *Naturalizing Phenomenology*, Stanford University Press — which treats conscious moments as emergent dynamics with genuine temporal extent (retention–primal impression–protention). Scribd **Evan Thompson's** *Mind in Life* (2007, Harvard University Press) treats consciousness as constitutively dependent on temporally extended processes. PhilPapers Notre Dame Philosophical ... **Tim van Gelder** (1998, "The Dynamical Hypothesis in Cognitive Science," *Behavioral and Brain Sciences* 21(5)) models cognition as state-space trajectories, though his claim is methodological rather than ontological. PhilPapers **Whitehead's** *Process and Reality* (1929) treats reality as composed of temporal "actual occasions" rather than substances, but these are brief atomic events, Wikipedia not trajectories in the dynamical-systems sense. Internet Encyclopedia of P... **IIT** (Oizumi, Albantakis, & Tononi, 2014, *PLoS Computational Biology* 10(5), e1003588) is fundamentally state-based, Internet Encyclopedia of P... evaluating cause-effect structure at single time-steps. Scholarpedia +4

(b) Closeness. DET reaches approximately **65% overlap**: it shares the ontological identification of consciousness with trajectories rather than states. However, DET's trajectories are generated step-by-step by forward-causal differential equations. Consciousnessrealist As Freeman explicitly noted, his K-set models show that "the system's state depends on its current position in state space, not on its past trajectory." ResearchGate No existing work proposes a **global self-consistency constraint** that selects admissible present states from whole-trajectory coherence.

(c) What remains novel. The self-consistency constraint across extended temporal intervals — the idea that the present is admissible only as part of a globally consistent trajectory, not generated causally step-by-step — is absent from all surveyed literature. This transforms the trajectory idea from a descriptive framework into a constitutive selection mechanism.

Claim 2: Binding works through temporal closure, not synchronization

The claim. Conscious unity arises because the system's present state is admissible only as part of a globally self-consistent trajectory. Present states are "selected" by whole-trajectory consistency, not generated by synchronization, recurrence, or global broadcast.

(a) Closest existing work. The binding problem has four major existing solutions, Wikipedia none of which invoke trajectory-level consistency:

- **Gamma synchrony**: Singer, W. & Gray, C.M. (1995), "Visual Feature Integration and the Temporal Correlation Hypothesis," *Annual Review of Neuroscience* 18, 555–586. Binding via millisecond-scale oscillatory synchronization — purely forward-causal. PubMed Central
- **Global Workspace Theory**: Baars, B.J. (1988), *A Cognitive Theory of Consciousness*, Cambridge University Press; Ucla Dehaene, S. & Naccache, L. (2001), *Cognition* 79(1–2), 1–37. Binding via broadcast to a global workspace — forward-causal information sharing. PubMed Central Theoriesofconsciousness
- **IIT**: Tononi, G. (2004), *BMC Neuroscience* 5, 42. Binding via integration (irreducibility of cause-effect structure) — state-based, not trajectory-based. Psychology Today +2
- **Recurrent Processing Theory**: Lamme, V.A.F. & Roelfsema, P.R. (2000), *Trends in Neurosciences* 23(11), 571–579. Binding via feedback loops — forward-causal temporal processing. PubMed Central Neuron

The **free energy principle** (Friston, K., 2010, *Nature Reviews Neuroscience* 11, 127–138) minimizes a functional over paths and has Lagrangian-like features, Frontiers but this optimization is performed moment-by-moment via gradient flows, Wikipedia UAB Medicine not through global trajectory constraints. **Hopfield networks** (Hopfield, J.J., 1982, *PNAS* 79(8), 2554–2558) implement constraint satisfaction, but spatially across units at a single time, converging to fixed-point attractors via forward dynamics. Wikipedia

(b) Closeness. No published work exceeds ~25% overlap. The specific proposal that binding works via whole-trajectory self-consistency — where present states are selected rather than generated — has no direct precedent. The Qualia Research Institute wiki contains a suggestive statement that "consciousness may be best viewed as a constraint satisfaction problem," but this lacks the temporal/trajectory dimension and mathematical formalization. Qri

(c) What remains novel. This appears to be **genuinely novel** in the consciousness literature. Every existing binding mechanism is forward-causal (synchronization, broadcast, recurrence, integration). The idea that binding is achieved by a global temporal consistency constraint that selects admissible present states from future-and-past coherence requirements has no published analogue. The closest structural relatives exist in physics (least action principle, Wheeler-Feynman absorber theory, consistent histories), not in consciousness studies.

Claim 3: Temporal mereology replaces spatial mereology for conscious subjects

The claim. The composition problem — which collections form a conscious subject — is solved by defining the subject as the maximal set of degrees of freedom jointly satisfying temporal closure, replacing spatial part-whole relations with trajectory-level co-participation.

(a) Closest existing work. The combination problem itself is well-defined by **Chalmers, D.J.** (2016), "The Combination Problem for Panpsychism," in Brüntrup & Jaskolla (eds.), *Panpsychism: Contemporary Perspectives*, Oxford University Press; Consc Oxford Academic and comprehensively treated by **Roelofs, L.** (2019), *Combining Minds*, Oxford University Press — but both operate within spatial/structural mereology.

Information Closure Theory (Chang, A.Y.C., Biehl, M., Yen, Y. & Kanai, R., 2020, "Information Closure Theory of Consciousness," *Frontiers in Psychology* 11, 1504) is the closest published framework. ICT defines conscious processes as those forming "non-trivial informational closure" at a particular coarse-grained scale arXiv — a temporal process whose future states are best predicted by its own past rather than by external variables. Medium ICT explicitly individuates conscious entities as processes, not spatial systems. PubMed Central

Van Inwagen, P. (1990), *Material Beings*, Cornell University Press, proposes that composition occurs iff activities constitute "a life" — an event-based, temporally defined criterion. ResearchGate Wikipedia **Whitehead's** actual occasions form "societies" through temporal ordering and mutual prehension, offering a processual mereology. Wikipedia Internet Encyclopedia of P... **McQueen, K.J. & Tsuchiya, N.** (2023), "When do parts form wholes?" *Neuroscience of Consciousness* 2023(1), niad013, directly use IIT's Φ as the answer to the Special Composition Question but remain within IIT's one-time-step framework.

(b) Closeness. ICT reaches approximately **45–50% overlap** and Whitehead reaches **40–45%**. ICT defines consciousness through temporal closure of processes, but its closure is statistical

(informational predictability), not trajectory-level non-factorizability. Whitehead provides temporal composition but lacks mathematical formalization in dynamical-systems terms. SciSpace Wikipedia No existing work explicitly **proposes replacing** spatial mereology with temporal mereology for consciousness.

(c) What remains novel. Three elements: the explicit framing of temporal mereology as a **replacement** for spatial mereology (not a supplement); the specific criterion of joint satisfaction of **temporal closure** as defining subject boundaries; and the concept of **trajectory-level co-participation** — degrees of freedom earn membership in a subject by contributing to a shared trajectory, not by occupying the same spatial region.

Claim 4: Non-factorizability of trajectories as the consciousness criterion

The claim. A system is conscious iff the whole trajectory satisfies a closure condition that no proper subset of its components can independently satisfy. Distinct from IIT's Φ because it is evaluated over temporal intervals, not instantaneous causal structure.

(a) Closest existing work. IIT's Φ (Oizumi et al., 2014, *PLoS Computational Biology*; Albantakis et al., 2023, "IIT 4.0," *PLoS Computational Biology* 19(10), e1011465) captures irreducibility — the whole generates more information than its parts — but over one-time-step transition probability matrices, not extended trajectories. PubMed Central +2

Kleiner, J. & Tull, S. (2020), "The Mathematical Structure of Integrated Information Theory," *Frontiers in Applied Mathematics and Statistics* 6, 602973, provide an axiomatic generalization of IIT that allows defining cause-effect repertoires "in reference to a given trajectory of a physical state" rather than requiring discrete Markovian dynamics. This is the closest formal precedent, reaching approximately **45–50% overlap**, but it generalizes IIT's existing framework rather than proposing trajectory non-factorizability as a competing criterion.

ICT (Chang et al., 2020) reaches similar overlap with its temporal closure criterion. arXiv **Oizumi & Tsuchiya** (2017, arXiv:1701.05157) extend IIT to continuous dynamics using delay embedding and introduce "temporal partitioning" but still measure integration at particular states or attractors. arXiv **Hoel, E.P.** (2013, *PNAS* 110(49)) defines causal emergence as a dimensionless quantity comparing macro vs. micro descriptions, PubMed sharing the architecture of "whole exceeds parts" but on the spatial-scale axis rather than the temporal one. **Kauffman's** catalytic closure (2000, *Investigations*, Oxford University Press) and **autopoietic** operational closure (Maturana & Varela, 1980) provide closure concepts that are temporal and holistic but lack formalization as trajectory non-factorizability and are not specifically about consciousness. ResearchGate

(b) Closeness. The formal ingredients — non-factorizability, trajectory analysis, closure conditions, integration as irreducibility — all exist. ICT and generalized IIT each share ~**50%**

conceptual territory. The gap is that no published work defines consciousness biconditionally via trajectory-level non-factorizability as a standalone criterion explicitly contrasted with instantaneous Φ.

(c) What remains novel. The specific synthesis: a **biconditional** criterion (iff) grounded in the non-factorizability of whole trajectories, with the formal object of analysis being extended temporal intervals rather than single-step transitions. This is more than a temporal extension of IIT — it changes what is being evaluated (trajectories, not states) and what property is tested (non-factorizability of the trajectory, not partition-based information loss at a time-step).

Claim 5: Time-symmetric, boundary-conditioned descriptions are structurally necessary

The claim. Drawing on Wheeler-Feynman absorber theory and Niknejadi-Madey-Kowalczyk (2015) on free-electron laser coherent radiation, some highly coupled, nonlinear, recursive systems cannot be naturally described by forward-only dynamics and require boundary-conditioned, effectively time-symmetric formulations. This applies to consciousness.

(a) Closest existing work. The physics foundations are well-established:

- **Wheeler, J.A. & Feynman, R.P.** (1945), *Reviews of Modern Physics* 17, 157–181; (1949), *Reviews of Modern Physics* 21, 425–434. Time-symmetric electrodynamics with advanced and retarded potentials; coherent radiation requires the absorber boundary condition. Wikipedia
- **Niknejadi, P., Madey, J.M.J. & Kowalczyk, J.M.D.** (2015), "Radiated power and radiation reaction forces of coherently oscillating charged particles in classical electrodynamics," *Physical Review D* 91, 096006. Demonstrates that standard retarded-only electrodynamics is "manifestly incompatible with Maxwell's equations' energy integral as applied to the process of coherent emission." The Wheeler-Feynman treatment provides a "perfect match." ResearchGate
- **Wharton, K.B.** (2013), "Lagrangian-Only Quantum Theory," arXiv:1301.7012, argues quantum mechanics should be understood via four-dimensional boundary-value problems, not initial-value problems. arXiv Systems are "solved all at once" with past and future constraints. San Jose State University
- **Aharonov, Y., Bergmann, P.G. & Lebowitz, J.L.** (1964), *Physical Review B* 134, 1410. The two-state vector formalism requires both initial and final boundary conditions for complete quantum description. Wikipedia
- **Price, H.** (1996), *Time's Arrow and Archimedes' Point*, Oxford University Press; Leifer, M. & Pusey, M. (2017), *Proceedings of the Royal Society A*. Time-symmetry plus quantum

discreteness implies retrocausality. Phys.org

- **Cramer, J.G.** (1986), *Reviews of Modern Physics* 58(3), 647–688. Transactional Interpretation with offer/confirmation wave handshakes. ResearchGate **Kastner, R.E.** (2012/2022), *The Transactional Interpretation of Quantum Mechanics*, Cambridge University Press, extends this but explicitly states TI "does not need to talk about consciousness." PhilPapers

Separately, **Vitiello, G.** (2001), *My Double Unveiled*, John Benjamins, applies quantum field theory to brain dynamics with dissipative mechanisms, but emphasizes time-asymmetry. **Stapp, H.P.** (2007), *Mindful Universe*, Springer, maintains forward-causal quantum collapse for consciousness. Wikipedia

(b) Closeness. The physics of structural necessity for boundary-conditioned descriptions is **well-proven** for specific electromagnetic systems (Niknejadi et al. 2015 being the clearest modern demonstration). Wharton and the TSVF provide the strongest arguments that quantum descriptions fundamentally require both initial and final boundary conditions. Stanford Encyclopedia of P... Springer However, **no published work bridges this to consciousness**. No one has treated neural/brain dynamics as a boundary-value problem. Searches for "boundary value neural dynamics" or "teleological neural computation" returned only technical papers using neural networks to solve mathematical BVPs, not papers modeling brains via boundary conditions.

A small speculative literature on retrocausal consciousness exists (Mossbridge et al., 2012, *Frontiers in Psychology*; Pitkänen's Topological Geometrodynamics), but these are either non-peer-reviewed, fringe, or don't make the structural-necessity argument.

(c) What remains novel. Four elements are genuinely new: the specific argument that consciousness, as a highly coupled nonlinear recursive system, **structurally requires** (not merely permits) boundary-conditioned descriptions; the use of the Niknejadi-Madey-Kowalczyk FEL result as a template for why such descriptions are necessary; the bridge from established physics results about coherent systems to neuroscience; and the treatment of brain dynamics as a boundary-value problem. The physics is real and well-cited. The bridge to consciousness is entirely original.

Claim 6: The Closure Dominance Parameter Ω

The claim. A dimensionless parameter $\Omega = 1 - (E_{RC} / E_R)$ measuring how much a system's intelligibility improves when temporal closure constraints are added versus retarded-only (forward-causal) modeling.

(a) Closest existing work. No dimensionless parameter specifically comparing forward-causal versus boundary-constrained descriptions was found in any domain. The closest structural analogues are:

- **Hoel's Causal Emergence**: CE = EI_macro − EI_micro (Hoel, E.P., Albantakis, L. & Tononi, G., 2013, *PNAS* 110(49), 19790–19795). Shares the architecture of a dimensionless quantity measuring how much a system's description improves under a different framework, but compares spatial scales (micro vs. macro), not temporal description types.
- **Bayesian model evidence** in Dynamic Causal Modeling (Friston, K.J. et al., 2003, *NeuroImage* 19, 1273–1302). Log Bayes factors quantify how much one model improves over another. UCL DCM has compared forward-only vs. forward-plus-backward connectivity models, but "forward/backward" here refers to hierarchical direction, not temporal direction. ScienceDirect
- **Kalman smoother vs. filter error ratio**: Kalman smoothing (using future data) always improves estimation over Kalman filtering (past-only), and the ratio of smoothing to filtering error is a well-defined quantity — but no standard named parameter exists for this ratio, and it has never been applied to consciousness.

IIT's **Φ** is a dimensionless consciousness measure but quantifies integration, not the improvement from switching description types. PubMed Central PubMed

(b) Closeness. Hoel's CE reaches approximately **30–35% structural overlap** — same conceptual architecture, different axis of comparison. The Bayesian model evidence framework provides the mathematical machinery to construct something like Ω but has never been applied this way. **No existing parameter directly measures what Ω measures.**

(c) What remains novel. The parameter itself — its definition, name, and conceptual role — appears to be **entirely novel**. No published work proposes a dimensionless quantity comparing forward-causal versus boundary-constrained temporal descriptions of a system. The concept of "closure dominance" as a named property is absent from physics, neuroscience, and complex systems literature. This is perhaps the most straightforwardly novel element of the entire framework.

Claim 7: Anesthesia reinterpreted as closure collapse

The claim. Anesthesia collapses temporal closure while preserving local coherence — the system becomes locally valid but globally unconstrained.

(a) Closest existing work. The empirical phenomenon described — local neural activity preserved, global coordination lost — is **extensively documented**:

- **Lewis, L.D. et al.** (2012), "Rapid fragmentation of neuronal networks at the onset of propofol-induced unconsciousness," *PNAS* 109, E3377–E3386. This is the closest empirical match (~**85% overlap**). Lewis et al. found that propofol-induced unconsciousness preserves "small-scale (<4 mm) functional connectivity" while disrupting global connectivity, concluding that propofol causes "functional isolation of cortical regions while significant connectivity is preserved within local networks."
- **Mashour, G.A.** (2004), "Consciousness unbound," *Anesthesiology* 100(2), 428–433; (2013), *Neuroscience & Biobehavioral Reviews* 37(10), 2751–2759. Mashour's "cognitive unbinding" hypothesis — that anesthesia disrupts the synthesis of functionally specialized modules while preserving local function — directly parallels "locally valid but globally unconstrained." NCBI
- **Alkire, M.T., Hudetz, A.G. & Tononi, G.** (2008), "Consciousness and Anesthesia," *Science* 322(5903), 876–880. IIT's framing: anesthesia causes loss of integration (Φ collapse) or loss of information capacity.
- **Casali, A.G. et al.** (2013), *Science Translational Medicine* 5(198), 198ra105. The Perturbational Complexity Index shows TMS-evoked responses become local and stereotyped under anesthesia. PubMed Frontiers
- **Hudetz, A.G.** (2012), *Brain Connectivity* 2(6), 291–302; **Mashour & Hudetz** (2024), *Nature Communications*. The integration-segregation difference (ISD) metric shows a "profound shift towards segregation" during anesthesia. Nature

(b) Closeness. This claim has the **highest overlap** with existing literature of all eight claims (~**75–85%**). The empirical observation (local preservation + global disruption) is well-established. The theoretical interpretations — integration loss (Tononi), cognitive unbinding (Mashour), PubMed cortical disconnection, complexity collapse (PCI), PLOS effective connectivity breakdown (Ferrarelli et al., 2010, *PNAS* 107(6)) — all describe essentially the same phenomenon.

(c) What remains novel. The novelty is primarily **terminological and conceptual**, not empirical. "Closure collapse" reframes existing observations in the vocabulary of temporal trajectory dynamics: what is lost is specifically the system's ability to maintain self-constraining trajectories over time (temporal closure), not just connectivity, integration, or complexity. The distinction between "locally valid" and "globally unconstrained" adds a dynamical-systems dimension absent from integration or unbinding language. But the underlying insight — that anesthesia preserves local function while dissolving global coherence PubMed Central — is well-known. **This is a re-description, not a discovery.**

Claim 8: Consciousness as entry into and exit from a closure regime

The claim. Consciousness emerges when a system enters a closure regime and disappears when that regime collapses. Minds are "drawn from" and "returned to" the space of unconstrained trajectories.

(a) Closest existing work. Multiple frameworks describe consciousness as a specific dynamical regime:

- **Critical brain hypothesis**: Chialvo, D.R. (2010), "Emergent complex neural dynamics," *Nature Physics* 6(10), 744–750; Beggs, J.M. & Plenz, D. (2003), *Journal of Neuroscience* 23(35), 11167–11177. Consciousness is associated with proximity to a critical point; departure from criticality → unconsciousness.
- **Comolatti, R. et al.** (2024), "Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness," *Communications Biology*. States of unconsciousness are "characterized by a distancing from both avalanche criticality and the edge of chaos." Nature
- **Global Workspace Theory** ignition: Dehaene, S. et al. (2021), *Neuron* 109(12), 1968–1983. Conscious access occurs when information crosses an ignition threshold, entering a self-sustaining broadcast mode.
- **Li, Z. et al.** (2025), "Falling asleep follows a predictable bifurcation dynamic," *Nature Neuroscience*. The wake-to-sleep transition follows bifurcation dynamics with a tipping point preceded by critical slowing down.
- **Solms, M.** (2019), *Frontiers in Psychology* 9, 2714, via Friston's FEP: self-organizing systems must resist "being dispersed over all possible states" — consciousness requires maintaining constrained states versus dispersing. PubMed Central
- **Carhart-Harris, R.L. et al.** (2014), "The entropic brain," *Frontiers in Human Neuroscience* 8, 20. Consciousness occupies a specific range on an entropy spectrum.

(b) Closeness. The general idea that consciousness corresponds to a specific dynamical regime with entry/exit transitions has **~55–65% overlap** with the criticality and phase-transition literatures. GWT's ignition model reaches ~55% with its threshold-crossing mechanism. The FEP's framing of self-organization as resistance to dispersion partially parallels "drawn from unconstrained trajectories."

(c) What remains novel. Three elements are new. First, the specific **"noise floor" metaphor** — unconstrained trajectories as a default background from which consciousness is drawn and to which it returns — has no published precedent. Second, the identification of the **regime-defining property as temporal closure** (trajectory self-constraint), rather than criticality, integration, or broadcast capacity, is distinct from all existing frameworks. Third, the

bidirectional symmetry of the framing — "drawn from and returned to" — treats emergence and dissolution as movements within the same space, rather than as separate phenomena requiring separate explanations. The core intuition (consciousness = dynamical regime) is well-established; the specific characterization of that regime is new.

What the framework actually contributes

Across all eight claims, a consistent pattern emerges. White Accumulation draws on well-established ingredients — dynamical systems theory of cognition (van Gelder, Freeman), PhilPapers California State University, ... trajectory-based consciousness (Fekete & Edelman), PhilPapers Consciousnessrealist temporal process ontology (Whitehead, Varela), Wikipedia Internet Encyclopedia of P... integrated information theory (Tononi), PubMed +4 time-symmetric physics (Wheeler-Feynman, Grokipedia Wharton, Aharonov), Stanford Encyclopedia of P... and the neuroscience of anesthesia (Mashour, Lewis, Hudetz) — but combines them in a configuration that has no published precedent. The framework's **genuinely novel core** is a single structural idea applied across all eight claims: that consciousness is constituted by a **global self-consistency constraint on temporally extended trajectories**, where present states are selected by whole-trajectory coherence rather than generated step-by-step. This idea — temporal closure as both the binding mechanism and the composition criterion — has no direct analogue in the surveyed literature.

The **strongest existing overlap** is with Information Closure Theory (Chang et al., 2020), which shares the temporal-process ontology and closure-based individuation but uses statistical predictability rather than trajectory non-factorizability. PubMed Central arXiv Generalized IIT (Kleiner & Tull, 2020) allows trajectory-based formulations within IIT's framework but does not propose them as a competing criterion. The **weakest novelty** is in Claim 7 (anesthesia), where the empirical content is well-known and the contribution is primarily a re-description. The **strongest novelty** resides in Claims 2 (binding via trajectory consistency), 5 (structural necessity of boundary-conditioned descriptions for consciousness), and 6 (the Ω parameter), where no close precedent was found in any surveyed domain.

The framework's deepest bet — that the move from initial-value to boundary-value formulations in physics (Wheeler-Feynman, Wharton, TSVF) Stanford Encyclopedia of P... should be replicated for consciousness — is the claim most likely to face resistance, since no one has modeled neural dynamics as a boundary-value problem. But it is also the claim that, if correct, would most fundamentally change how consciousness is studied. The physics cited is real and peer-reviewed. The bridge to consciousness is where all the risk — and all the originality — resides.